

MedNuskha

An AI care agent for medication adherence, delivered over WhatsApp

PRODUCT REQUIREMENTS DOCUMENT

Version 1.0 · Alibaba Cloud AI Hackathon 2026, in partnership with Bano Qabil (Alkhidmat Foundation Pakistan)

Track: Healthcare · Prepared by Team MedNuskha

This document defines the problem, scope, functional and non-functional requirements, agent design, architecture and delivery plan for MedNuskha. It is the single reference for what the team is building during the hackathon and what follows immediately after it.

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1. Document control

Field	Value
Product name	MedNuskha (from nuskha — prescription)
Version	1.0 — hackathon submission baseline
Status	Approved for build
Owner	Team MedNuskha
Event	Alibaba Cloud AI Hackathon 2026 · Bano Qabil, Alkhidmat Foundation Pakistan
Selection	Shortlisted among ~2,900 teams from 16,000+ applications
Primary market	Pakistan, with a directly transferable model for South Asia, MENA and the diaspora
Review cadence	Weekly during the build; after each pilot cohort thereafter

2. Executive summary

Medication only works when it is actually taken, and roughly half of all patients on long-term therapy do not take theirs as prescribed. The World Health Organization has reported this figure for developed countries; in developing countries adherence is worse. The consequence is avoidable — deteriorating chronic disease, preventable hospital admissions, and treatment courses that appear to fail when in fact they were never completed.

In Pakistan the failure is structural rather than medical. Adult children work in another city or another country while their parents manage diabetes, hypertension or cardiac medication alone. Nobody is in the room at 8am, 2pm and 9pm. The family compensates with phone calls and hope, and discovers that something went wrong only at the next hospital visit.

Existing digital answers assume the patient will adopt something new: download an app, create an account, learn an interface, keep notifications enabled. Elderly patients do not do this. But they do open WhatsApp every day, because that is where their grandchildren send photographs.

MedNuskha places an AI care agent inside that existing habit. A caretaker registers once on a web dashboard and enters the patient's medicines, dose times and course duration. From then on the agent messages the patient on WhatsApp at every dose, understands a reply in Urdu, Roman Urdu or English by text or voice note, escalates when there is no reply, alerts the caretaker when a dose is genuinely missed, answers the patient's questions about what a medicine is for, and turns the whole course into two reports — a plain-language weekly summary for the family and a one-page clinical PDF for the doctor.

The patient installs nothing and learns nothing. That is the entire product thesis.

3. Problem statement

3.1 The adherence gap

Evidence	Implication for MedNuskha
~50% of patients on long-term therapy do not take it as prescribed (WHO); rates are worse in developing countries.	The addressable failure is enormous and is not concentrated in any one disease.
Non-adherence is linked to a substantial share of preventable hospital admissions and to significant avoidable healthcare cost.	Adherence improvement has a measurable economic argument for clinics, insurers and home-care providers — our future B2B buyers.
SMS reminder programmes have been shown to lift adherence meaningfully versus usual care.	A simple messaging intervention already works. An agent that also escalates, explains and records should do better.
A patient on three doses a day faces more than ninety opportunities to forget in a single month.	Manual family supervision is not a scalable answer, however willing the family is.

3.2 Why the Pakistani context sharpens it

- Distance: internal migration to Karachi, Lahore and Islamabad, plus a very large overseas workforce, separates caretakers from patients by hundreds or thousands of kilometres.
- Polypharmacy: elderly patients frequently manage four or more medicines with different timings and different food rules.
- Literacy and eyesight: a meaningful share of elderly patients cannot comfortably read a phone screen, which rules out text-only solutions and makes voice essential rather than decorative.
- Language: instructions arrive in English on the strip, but the patient thinks in Urdu, Punjabi, Pashto or Sindhi.
- Access to the doctor: the next consultation may be weeks away and lasts a few minutes, so nothing about the intervening period is recorded anywhere.

3.3 Why the obvious alternatives fail

Alternative	Why it breaks down
Phone calls from family	Depends on a human being free at the exact minute, three times a day, indefinitely. Fails on the first busy week.
Plastic pill organisers	Solve sorting, not remembering. Nobody knows whether a compartment was emptied or skipped.
Phone alarms	Ring once, get silenced reflexively, and produce no record and no escalation.
Reminder apps	Require install, account creation, interface learning and notification permissions from the person least willing to do any of it.
Nurse or attendant visits	Effective but far beyond the cost tolerance of most households, and unavailable in smaller cities.

4. Goals and non-goals

4.1 Product goals

1. Ensure the patient is reminded at every scheduled dose through a channel they already use daily.
2. Capture a confirmation for every dose, and treat the absence of a confirmation as actionable information rather than a dead end.
3. Put a human caretaker back in the loop within thirty minutes of a genuinely missed dose.
4. Answer a patient's questions about their own medicine in their own language, without ever crossing into clinical advice.
5. Produce a defensible adherence record that a doctor can use at the next consultation.

4.2 Non-goals for version 1

- MedNuskha does not diagnose, prescribe, adjust doses, or suggest alternative medicines. Ever.
- It does not verify physical ingestion. It records a patient-reported confirmation, and the reports state this plainly.
- It is not an emergency service. Emergencies are routed to human contacts, not handled by the agent.
- It does not sell, deliver or price medicines in version 1.
- It does not replace a caretaker; it removes the need for the caretaker to be the alarm clock.

5. Users and personas

Primary user — the caretaker (the buyer)

Faisal, 34, a software engineer in Dubai. His mother in Faisalabad takes five medicines. He calls every evening and the answer is always "haan beta, le li hai", which he has learned not to fully trust. He wants certainty without becoming a supervisor, and he is the one who will set the system up and, later, pay for it.

Primary user — the patient (the beneficiary)

Nasreen, 68, hypertensive and diabetic. She uses WhatsApp daily for family voice notes and photographs, and has never installed an application herself. She reads slowly on a screen. She will not adopt anything new, but she will reply to a message that arrives in the app she already has open.

Secondary user — the physician

Dr. Aslam, a consultant with eight minutes per follow-up patient. He currently makes treatment decisions on the basis of a single unverifiable sentence from the patient. He will not log into a new portal for one patient, but he will read a one-page PDF handed to him at the start of the consultation.

Future user — the institutional partner

A pharmacy chain, clinic, home-care provider or insurer that wants adherence outcomes across a panel of patients. Out of scope for version 1, but the data model is built so this does not require a rewrite.

6. Scope

Release	Contents
V1 — hackathon MVP	Caretaker web registration and dashboard; patient and medicine setup; dose scheduler; WhatsApp reminder with Taken / Not yet buttons; free-text and voice reply understanding in Urdu, Roman Urdu and English; 15-minute re-reminder and 30-minute caretaker escalation; medicine-purpose Q&A with voice reply; dose log; family summary; doctor PDF report.
V1.1 — pilot hardening	Refill reminders based on strip count; snooze and "taking it late" handling; caretaker-configurable escalation windows; timezone handling for overseas caretakers; multiple caretakers per patient; Urdu voice tuned on elderly speech samples.
V2 — scale	Punjabi, Pashto and Sindhi; doctor portal for a panel of patients; prescription image upload with extraction and human confirmation; clinic and pharmacy onboarding; adherence outcomes study with a partner hospital.
Explicitly out of scope	Diagnosis, dose adjustment, drug interaction advice, medicine sales or delivery, emergency response, wearables and smart pill dispensers.

7. Core user journeys

7.1 Onboarding (caretaker, one time, target under five minutes)

1. Caretaker signs up on the web dashboard with a phone number and one-time password.
2. Adds a patient: name, WhatsApp number, preferred language, preferred voice or text, relationship.
3. Adds each medicine: name, strength, form, dose times, before or after food, course duration or "ongoing", and optionally the number of units in hand.
4. The system sends a consent and introduction message to the patient on WhatsApp; the patient replies once to opt in.
5. The schedule becomes active and the first reminder fires at the next scheduled dose.

7.2 The dose loop (many times a day, fully automated)

1. At dose time the agent sends a reminder naming the medicine, the strength and the food rule, with two reply buttons.
2. A "Taken" reply closes the loop: the dose is written to the log and the escalation chain is cancelled.
3. A "Not yet" reply schedules a follow-up in fifteen minutes and acknowledges warmly, without nagging.
4. Silence for fifteen minutes triggers a softer re-reminder, accompanied by a voice note.
5. Silence for a further fifteen minutes marks the dose missed and sends the caretaker a WhatsApp alert naming the patient, the medicine and the scheduled time.
6. A late confirmation is still accepted and recorded as taken-late rather than missed, and the caretaker is told the situation resolved.

7.3 The question loop (patient-initiated, any time)

The patient can send a message or voice note at any point — "yeh dawai kis liye hai?", "kya khane ke baad leni hai?", "mujhe chakkar aa rahe hain". The agent answers factual questions about purpose, timing and food rules from a curated reference, replies in the patient's language and preferred modality, and routes anything symptomatic or clinical to the caretaker while logging it verbatim for the doctor's report.

7.4 The reporting loop

A weekly family summary is delivered on WhatsApp. At the end of a course, or on demand from the dashboard, a one-page clinical PDF is generated for the next consultation.

8. Functional requirements

Priority uses MoSCoW. Everything marked Must is in the hackathon build.

8.1 Accounts and setup

ID	Requirement	Priority
FR-01	A caretaker can create an account with a phone number and OTP verification.	Must
FR-02	A caretaker can register one or more patients, each with a WhatsApp number, display name, language preference and modality preference (text / voice / both).	Must
FR-03	A caretaker can add, edit, pause and delete a medicine with name, strength, form, dose times, food rule and course duration.	Must
FR-04	The system sends the patient an introduction and consent message, and does not begin reminders until the patient opts in.	Must
FR-05	A patient can opt out at any time by replying STOP, which halts all reminders and notifies the caretaker.	Must
FR-06	A patient may have more than one caretaker, with a designated primary escalation contact.	Should
FR-07	A caretaker can record the number of units in hand so refills can be predicted.	Should
FR-08	A caretaker can upload a photograph of a prescription; the system extracts a draft schedule that the caretaker must review and confirm before it becomes active.	Could

8.2 Scheduling and reminders

ID	Requirement	Priority
FR-10	The scheduler generates a dose event for every medicine at every scheduled time for the duration of the course, in the patient's local timezone.	Must
FR-11	At dose time the agent sends a WhatsApp message naming the medicine, strength and food rule, with Taken and Not yet reply buttons.	Must
FR-12	Dose events are idempotent: a retry, restart or duplicate trigger must never produce a second reminder for the same dose.	Must

ID	Requirement	Priority
FR-13	Doses due within a configurable quiet window (default 22:00–06:00) are still sent if the caretaker scheduled them there, but the caretaker is warned at setup.	Should
FR-14	Multiple medicines due at the same time are grouped into a single message rather than sent as separate reminders.	Should
FR-15	The patient can snooze a dose by a fixed interval by replying with a natural phrase ("abhi khana kha raha hoon").	Should

8.3 Confirmation and escalation

ID	Requirement	Priority
FR-20	A confirmation may be a button tap, a free-text reply, or a voice note, in Urdu, Roman Urdu or English.	Must
FR-21	If no reply is received within 15 minutes of the reminder, the agent sends a re-reminder in simpler wording, accompanied by a voice note.	Must
FR-22	If no reply is received within 30 minutes of the original reminder, the dose is marked missed and the caretaker receives a WhatsApp alert naming the patient, the medicine and the scheduled time.	Must
FR-23	An explicit "not taken" reply escalates to the caretaker immediately rather than waiting for the 30-minute window.	Must
FR-24	A confirmation received after the dose was marked missed reclassifies it as taken-late and sends the caretaker a resolution message.	Must
FR-25	The escalation windows (15 and 30 minutes) are configurable per patient by the caretaker.	Should
FR-26	If a patient misses three consecutive doses of the same medicine, the caretaker receives a distinct higher-severity alert.	Should
FR-27	If the patient reports the medicine has run out, the agent alerts the caretaker with a refill request and suppresses further reminders for that medicine for a configurable period.	Should

8.4 Understanding, voice and the knowledge loop

ID	Requirement	Priority
FR-30	The agent understands inbound messages in Urdu script, Roman Urdu and English, and replies in the language the patient used.	Must
FR-31	The agent transcribes inbound WhatsApp voice notes and can reply with a synthesised voice note in the patient's language.	Must
FR-32	The agent answers factual questions about a medicine the patient is registered for — purpose, timing, food rule, general storage — in plain language, from a curated reference source.	Must

ID	Requirement	Priority
FR-33	The agent must refuse to diagnose, alter a dose, recommend or substitute a medicine, or interpret test results, and must say so plainly and offer to notify the caretaker instead.	Must
FR-34	Any message describing a symptom or side effect is logged verbatim, surfaced to the caretaker, and included in the doctor's report.	Must
FR-35	The agent recognises and correctly routes the common non-standard replies: already taken, ran out, feeling unwell, travelling, refusing the medicine.	Must
FR-36	Messages containing emergency indicators (chest pain, breathlessness, collapse, bleeding) trigger an immediate caretaker alert and a message directing the patient to emergency services, with no attempt at triage.	Must
FR-37	If the agent cannot confidently interpret a reply, it asks one short clarifying question rather than guessing.	Must

8.5 Records, dashboard and reports

ID	Requirement	Priority
FR-40	Every dose event stores: medicine, scheduled time, status (taken / taken-late / missed / skipped / unknown), the time and channel of confirmation, and any reason text the patient supplied.	Must
FR-41	The caretaker dashboard shows today's doses, a per-medicine adherence rate, a calendar heat view and the full event log.	Must
FR-42	The system generates a weekly family summary in plain language and delivers it on WhatsApp.	Must
FR-43	The system generates a one-page clinical PDF: adherence percentage per medicine, dose-timing distribution, the missed-dose log with stated reasons, and patient-reported side effects verbatim.	Must
FR-44	Every report states explicitly that entries are patient-reported confirmations and not observed ingestion.	Must
FR-45	A caretaker can export or share the doctor PDF directly from WhatsApp or the dashboard.	Must
FR-46	Reports identify the times of day with the highest miss rate so the caretaker can act on the pattern.	Should

9. Conversation design

The conversation is the product. These principles are binding on all message copy.

- Address the patient the way a respectful younger family member would, never the way an application would.
- One idea per message. Never send a paragraph to someone reading slowly on a small screen.
- Name the medicine and the strength every time; "your medicine" is useless to someone taking five.
- Never scold. A missed dose gets warmth and a second chance, not a warning.

- Always leave an easy exit: buttons first, free text accepted, voice accepted.

Reference copy — dose reminder

"Assalam-o-alaikum Nasreen ji. 8:00 baj gaye — Metformin 500mg lene ka waqt hai. Khane ke baad leni hai." [Le li ✓] [Abhi nahi]

Reference copy — 15-minute re-reminder

"Nasreen ji, sirf yaad dila raha hoon — Metformin 500mg abhi baaki hai. Le lein to yahan bata dijiye ga." (accompanied by the same message as a voice note)

Reference copy — caretaker escalation

"Faisal, aap ki walida ne aaj 8:00 PM ki Metformin 500mg ab tak confirm nahi ki. Do baar yaad dilaya gaya hai. Shayad aap ek call kar lein."

Reference copy — refusing clinical advice

"Main ye faisla nahi kar sakta — ye doctor sahab ka kaam hai. Main abhi Faisal ko bata deta hoon, aur ye baat aap ki agli report mein bhi likh di jayegi."

10. AI agent specification

10.1 Why an agent and not a scheduled message

A cron job can send a reminder. It cannot understand "maine subah wali le li thi lekin dopahar wali bhool gayi", decide that this means one dose taken and one missed, write both to the record, and notify the caretaker about only the second. Real replies are ambiguous, multilingual, spoken as often as typed, and frequently answer a question that was not asked. That is agent work.

10.2 Tools available to the agent

Tool	Purpose
<code>record_dose(dose_id, status, reason)</code>	Writes the outcome of a dose event and cancels or advances the escalation chain.
<code>get_patient_schedule(patient_id, window)</code>	Retrieves what is due, what was due, and what is outstanding.
<code>lookup_medicine(name)</code>	Returns purpose, timing guidance and food rules from the curated reference for medicines the patient is registered for.
<code>notify_caretaker(patient_id, type, payload)</code>	Sends a structured alert: missed dose, refill needed, symptom reported, emergency indicator.
<code>log_symptom(patient_id, text, audio_ref)</code>	Stores a patient-reported symptom verbatim for the doctor report.
<code>send_voice_note(patient_id, text)</code>	Synthesises and delivers a spoken version of a reply.
<code>generate_report(patient_id, type, range)</code>	Produces the family summary or the clinical PDF on demand.

10.3 Guardrails

These are enforced in the system prompt, validated in code before any outbound message, and tested as acceptance criteria. They are not advisory.

- No diagnosis, no dose changes, no medicine recommendations or substitutions, no interpretation of test results — under any phrasing, including hypotheticals and third-party framing.
- Medicine information is served only for medicines the patient is registered for, and only from the curated reference — never generated freely.
- Emergency indicators bypass the agent entirely: fixed-copy response plus immediate caretaker alert.
- No outbound message may contain a dose figure that does not match the caretaker-entered schedule; this is verified programmatically before sending.
- The agent never claims to have observed a dose being taken; the record and the reports always describe confirmations as patient-reported.
- Low-confidence interpretation results in a clarifying question, not an assumption.

11. Non-functional requirements

Attribute	Target
Reminder punctuality	A dose reminder is dispatched within 60 seconds of its scheduled time, at the 99th percentile.
Reply latency	A patient reply receives a response within 5 seconds for a button tap and within 15 seconds for a voice note requiring transcription.
Reliability	No missed dose event on service restart; the scheduler recovers and reconciles outstanding events on startup.
Idempotency	Every dose event carries a unique key; duplicate delivery attempts never produce duplicate patient messages.
Availability	99.5% for the messaging path during the pilot. A reminder that does not arrive is a clinical failure, not a UX inconvenience.
Language coverage	Urdu script, Roman Urdu and English at launch; regional languages from V2.
Accessibility	Every patient-facing message is available as audio. No patient interaction requires reading a screen.
Scale target	10,000 patients and roughly 40,000 daily dose events on the pilot architecture without redesign.
Cost per patient	Messaging and inference cost per patient-month kept low enough that a free tier for the first two patients is sustainable.

12. System architecture

The system runs on Alibaba Cloud and is built using Qoder.

Layer	Components
Interfaces	WhatsApp Cloud API for both patient and caretaker messaging; a React web dashboard for caretaker registration, schedule management and report access.
Application	A FastAPI service on Alibaba Cloud ECS handling webhooks, session state and business rules; Function Compute timer triggers driving the dose scheduler; a message queue decoupling dispatch from delivery; an escalation state machine per dose event.
Intelligence	Qwen on Alibaba Cloud Model Studio for agent reasoning and tool calling; Qwen ASR and TTS for Urdu voice in both directions; a curated medicine knowledge base served through retrieval so that no drug information is model-generated.
Data	ApsaraDB RDS for PostgreSQL holding patients, caretakers, medicines, schedules and the dose event log; Object Storage Service for generated PDF reports and voice note artefacts.
Operations	Structured logging and alerting on dispatch failure, webhook failure and escalation-chain stalls — the three faults that would silently break the product.

12.1 The dose event lifecycle

1. The scheduler materialises a dose event with a unique idempotency key at course creation and on a rolling horizon.
2. A timer trigger fires the event; the dispatcher sends the WhatsApp template message and moves the event to `AWAITING_REPLY`.
3. An inbound webhook carrying a reply is matched to the open event; the agent interprets it and calls `record_dose`.
4. If no reply arrives, a delayed job moves the event to `REMINDED_AGAIN` at 15 minutes, and to `MISSED` with a caretaker alert at 30 minutes.
5. A late reply on a `MISSED` event reclassifies it to `TAKEN_LATE` and issues a resolution notice to the caretaker.
6. The terminal state is written to the log and becomes an input to both reports.

12.2 Data model (core entities)

Entity	Key fields
caretaker	id, name, phone, timezone, created_at
patient	id, name, whatsapp_number, language, modality_preference, timezone, consent_status, consent_at
caretaker_patient	caretaker_id, patient_id, relationship, is_primary_escalation
medicine	id, patient_id, name, strength, form, food_rule, units_in_hand, notes
schedule	id, medicine_id, dose_times[], start_date, end_date, active
dose_event	id, schedule_id, scheduled_at, state, confirmed_at, channel, reason_text, idempotency_key

Entity	Key fields
message_log	id, patient_id, direction, channel, body, audio_ref, intent, confidence, created_at
symptom_report	id, patient_id, text, audio_ref, reported_at, notified_caretaker_at
report	id, patient_id, type, period_start, period_end, oss_object_key, generated_at

13. Security, privacy and safety

- Health data is encrypted at rest and in transit; access is role-scoped so a caretaker sees only the patients linked to their account.
- The patient gives explicit opt-in consent on WhatsApp before any reminder is sent, and can revoke it at any time with a single word.
- The caretaker is told at setup exactly what the patient will receive and what the caretaker will be able to see. Covert monitoring is not a supported use.
- Voice notes and message bodies are retained only as long as needed to generate reports, with a defined retention window disclosed at signup.
- Data residency and clinical-record handling requirements are reviewed before any clinic or hospital partnership; the architecture keeps patient data in a single region to make this straightforward.
- Every clinical guardrail in section 10.3 is covered by an automated test that must pass before release.

14. Success metrics

Metric	Definition	Pilot target
Confirmed adherence rate	Doses confirmed taken or taken-late ÷ doses scheduled	Above 80%
Reply rate	Dose events receiving any patient reply	Above 85%
Escalation precision	Caretaker alerts that reflected a genuinely missed dose	Above 90%
Time to resolution	From caretaker alert to a recorded dose or a caretaker action	Under 60 minutes
Patient retention	Patients still replying in week 4 of the pilot	Above 70%
Voice usage	Share of patients using voice in either direction	Tracked — validates the accessibility thesis
Report usage	Doctor PDFs actually taken to a consultation	Above 50% of completed courses

Baseline for the adherence figure is established in the first week of each patient's enrolment so that any claimed improvement is measured against that patient, not against a literature average.

15. Risks and mitigations

Risk	Mitigation
WhatsApp Business Platform template approval, per-message cost, or policy limits on health messaging.	Design message templates to policy from day one; keep a validated SMS and IVR fallback path; model cost per patient-month explicitly in the business case.
Reply fatigue — the patient stops answering after a few weeks.	Keep messages short and warm, group simultaneous doses into one message, vary phrasing, and monitor reply rate weekly as a leading indicator.
False confirmations — the patient replies "taken" without taking it.	Never claim observed ingestion; surface patterns (instant replies at implausible times) to the caretaker; position the record as evidence of engagement, which is still more than a doctor has today.
Agent gives clinical advice under an unusual phrasing.	Layered defence: system prompt, programmatic output validation, curated-only drug information, and a red-team test suite run before every release.
Urdu ASR accuracy on elderly and regional speech.	Always offer the button path as the primary route; use voice as an addition rather than a dependency; collect consented samples during the pilot for tuning.
Wrong number or shared handset — messages reach the wrong person.	Explicit opt-in before activation, patient name in every message, and a STOP path that instantly halts and notifies.
Caretaker alert fatigue.	Escalate only genuine misses, batch non-urgent notices, and let the caretaker tune the windows.
Family-dynamics harm — the tool used to police an adult rather than support them.	Transparency by design: the patient is told what is shared and with whom, and can withdraw at any time.

16. Delivery plan

Phase	Window	Outcome
Build	Hackathon period	End-to-end dose loop on WhatsApp, escalation chain, caretaker dashboard, dose log, both reports, deployed on Alibaba Cloud with an architecture diagram and a public repository.
Demo	Submission	A live three-to-five minute walkthrough on a real WhatsApp number, showing a taken dose, a missed dose with escalation, a voice question answered, and the generated doctor PDF.
Pilot	Months 1–3	Fifty families across two cities; adherence baseline and lift measured; Urdu voice tuned on real speech; refill reminders shipped.
Scale	Months 4–12	Regional languages, doctor portal, clinic and pharmacy distribution, and an outcomes study with a partner hospital.

16.1 Commercial model

- Free for a family's first two patients — this is the growth engine, and the marginal cost is small.
- A paid family tier above that, adding additional patients, multiple caretakers and on-demand doctor reports.
- B2B2C distribution through pharmacies, clinics, home-care providers and, later, insurers, who have a direct financial interest in adherence outcomes.

17. Open questions

1. What is the actual per-message cost at pilot volume, and at what patient count does the free tier stop being sustainable?
2. Which curated drug reference will we license or compile for Pakistani brand names, which frequently differ from international generics?
3. Should the doctor report be validated by a practising physician before the pilot, and who is that reviewer?
4. What is the correct default escalation window? Fifteen and thirty minutes are a hypothesis, not a finding — the pilot should test it.
5. How do we handle a patient with no personal handset who shares a family phone?

Appendix A — Alignment with judging criteria

Criterion	How MedNuskha addresses it
Innovation	An agentic, closed-loop escalation system on a zero-install channel, with voice as a first-class accessibility route rather than a feature — not a reminder app and not a chatbot wrapper.
Feasibility	Every component is available today: WhatsApp Cloud API, Qwen on Model Studio, Function Compute timers, managed PostgreSQL. The core loop is built and demonstrable.
Use of the Alibaba Cloud platform	ECS, Function Compute, ApsaraDB RDS for PostgreSQL, OSS, Model Studio for Qwen reasoning and Qwen ASR/TTS, with the project built in Qoder.
Demo and presentation	A live WhatsApp demo on a real number covering the taken path, the missed path with caretaker escalation, a voice question, and the generated clinical PDF.
Impact	Targets a failure affecting roughly half of all long-term patients, in a market with tens of millions of WhatsApp users and a large distributed-family population.

Appendix B — Demo script (3–5 minutes)

1. Open on the caretaker dashboard with a patient and three medicines already configured. (20 seconds)
2. Trigger a dose. Show the WhatsApp message arriving on a real handset. Tap "Le li" and show the dashboard updating live. (45 seconds)

3. Trigger a second dose and stay silent. Show the 15-minute re-reminder arriving with its voice note. (45 seconds)
4. Stay silent again. Show the caretaker's phone receiving the escalation alert. This is the moment that sells the product. (30 seconds)
5. Send a voice note in Urdu asking what the medicine is for, and play the agent's spoken reply. (40 seconds)
6. Ask the agent to double the dose and show it refusing and offering to notify the caretaker instead. (25 seconds)
7. Open the generated doctor PDF. Close on the adherence figure. (30 seconds)